1. You can use the “expr” command to calculate simple math.

```
kmin@kali-rrc:~$ expr 1 + 5
6
kmin@kali-rrc:~$ expr 1+5
1+5
kmin@kali-rrc:~$ expr 1 * 5
expr: syntax error: unexpected argument ‘bills_keys’
kmin@kali-rrc:~$
```

2. If you want to multiply, you may need to enter a \ to work.

```
kmin@kali-rrc:~$ expr 1 \* 5
5
kmin@kali-rrc:~$
```

3. You can use $( ) to enter the output of the command as the input in the terminal.

```
kmin@kali-rrc:~$ echo hello
hello
kmin@kali-rrc:~$ $(echo hello)
bash: hello: command not found
kmin@kali-rrc:~$ $(echo pwd)
/home/kmin
kmin@kali-rrc:~$ $(echo ls)
bills_keys      Desktop    file0example.txt
bills_keys.pub  Documents  file1example.txt
case.sh         Downloads  file2example.txt
dataGen.sh      elseif.sh  file3example.txt
kmin@kali-rrc:~$
```

4. Testing a bad command in if statement

```
kmin@kali-rrc:~/if_statements$ chmod +x test2.sh
kmin@kali-rrc:~/if_statements$ sudo ./test2.sh
./test2.sh: line 3: IamNotAcommand: command not found
We're out of the if statement
kmin@kali-rrc:~/if_statements$
```

5. test3

```
kmin@kali-rrc:~/if_statements$ chmod +x test3.sh
kmin@kali-rrc:~/if_statements$ sudo ./test3.sh
kmin:x:1000:1000:,,,:/home/kmin:/bin/bash
This is my first command
This is my second command
I can even put other commands besides echo
.               .cache      elseif.sh         file5example.txt
..              case.sh     .face             file6example.txt
.bash_history   .config     .face.icon        file7example.txt
.bash_logout    dataGen.sh  file0example.txt  file8example.txt
.bashrc         Desktop     file1example.txt  file9example.txt
.bashrc.original  .dmrc     file2example.txt  .gnupg
bills_keys      Documents   file3example.txt  .ICEauthority
bills_keys.pub  Downloads   file4example.txt  if_statements
kmin@kali-rrc:~/if_statements$
```

6. test4

```
kmin@kali-rrc:~/if_statements$ sudo ./test4.sh
The user NoSuchUser does not exist on the system

kmin@kali-rrc:~/if_statements$
```

7. Test5

```
kmin@kali-rrc:~/if_statements$ sudo ./test5.sh
The user NoSuchUser does not exist on the system
ls: cannot access '/home/NoSuchUser/': No such file or directory
kmin@kali-rrc:~/if_statements$ nano test5.sh
kmin@kali-rrc:~/if_statements$
```

8. Test6

```
kmin@kali-rrc:~/if_statements$ sudo ./test6.sh
[sudo] password for kmin:
The  expression returns a False
kmin@kali-rrc:~/if_statements$
```

9. Numeric_test

```
kmin@kali-rrc:~/if_statements$ sudo ./numeric_test.sh
The test value 10 is greater than 5
The values are different.
kmin@kali-rrc:~/if_statements$
```

10. Test7

```
kmin@kali-rrc:~/if_statements$ ./test7.sh
Welcome kmin
kmin@kali-rrc:~/if_statements$
```

11. Test8

```
kmin@kali-rrc:~/if_statements$ sudo ./test8.sh
This is not baduser
kmin@kali-rrc:~/if_statements$
```

12. Test9b

```
kmin@kali-rrc:~/if_statements$ sudo ./test9b.sh
val1 is less than val2
kmin@kali-rrc:~/if_statements$
```

13. Test13

```
kmin@kali-rrc:~/if_statements$ ./test13.sh

The item being checked: /home/kmin

The item, /home/kmin, does exist.
But, is it a file?

No, /home/kmin is not a file.
kmin@kali-rrc:~/if_statements$
```

14. Test14

```
kmin@kali-rrc:~/if_statements$ ./test14.sh
Sorry, I am unable to read /etc/shadow
kmin@kali-rrc:~/if_statements$
```

15. Test18

```
kmin@kali-rrc:~/if_statements$ ./test18.sh
Sorry, you are not the owner of the /etc/passwd file
kmin@kali-rrc:~/if_statements$
```

16. Test21

```
kmin@kali-rrc:~/if_statements$ ./test21.sh
Badfiel2 is newer than badfile1
kmin@kali-rrc:~/if_statements$
```

17. Test22

```
kmin@kali-rrc:~/if_statements$ ./test22.sh
I cannot write to the file
kmin@kali-rrc:~/if_statements$ touch $HOME/testing
kmin@kali-rrc:~/if_statements$ ./test22.sh
The file exists and you can write to it
kmin@kali-rrc:~/if_statements$
```

18. Test25

```
kmin@kali-rrc:~/if_statements$ ./test25.sh
Welcome kmin
Please enjoy your visit
kmin@kali-rrc:~/if_statements$
```

19. Test26

```
kmin@kali-rrc:~/if_statements$ ./test26.sh
Welcome kmin
Please enjoy your visit
kmin@kali-rrc:~/if_statements$
```